

Titan Assistant Architecture

Purpose

This document defines the assistant architecture used by TitanZero.

Assistants are the intelligent agents that monitor system activity, interpret signals, and propose actions to improve business operations.

Assistants operate under the governance of the Titan Zero Engine and interact with the system through signals and engine APIs.

They do not directly modify operational tables.

Instead, they propose actions that are executed through the Titan Pulse Engine or approved by governance workflows.

Assistant Philosophy

Titan assistants function as lifecycle observers and decision helpers.

They continuously monitor signals emitted by the system and assist in managing the lifecycle of entities such as:

- jobs
- quotes
- invoices
- campaigns
- customers
- documents

Assistants help the system operate proactively rather than reactively.

Assistant Types

TitanZero contains two primary assistant families.

LogiCore Assistants

CreatiCore Assistants

Each assistant type focuses on different business functions.

LogiCore Assistants

LogiCore assistants manage operational lifecycles.

They focus on structured business processes.

Examples include:

- job lifecycle assistants
- dispatch assistants
- invoice lifecycle assistants
- quality control assistants
- customer service assistants

Responsibilities include:

- monitoring job progress
- detecting operational problems
- triggering follow-up tasks
- managing scheduling and dispatch
- ensuring financial processes occur correctly

LogiCore assistants primarily respond to work, financial, and operational signals.

CreatiCore Assistants

CreatiCore assistants manage marketing and creative workflows.

They assist with activities such as:

campaign creation
content generation
audience targeting
promotion analysis
engagement optimization

Responsibilities include:

creating marketing campaigns
generating promotional messages
analysing campaign performance
adjusting marketing strategies

CreatiCore assistants respond primarily to customer and communication signals.

Assistant Activation

Assistants are activated when relevant signals occur.

Example activation:

work.job.completed

Assistant response:

LogiCore lifecycle assistant evaluates job completion

→ proposes invoice creation
→ proposes review request
→ proposes next service scheduling

Assistants interpret the signal and generate suggested actions.

Assistant Proposals

Assistants generate structured proposals describing recommended actions.

Example proposal:

proposal_type: action

action:

create_invoice_from_job

context:

job_id: 4829

customer_id: 14

service_type: cleaning

Proposals are submitted to Titan Pulse or Titan Zero governance for execution.

Assistant Routing

Assistants are managed through the Titan Zero Engine router.

The router selects the appropriate assistant based on:

signal type

entity type

business context

risk level

Example routing:

Signal: work.job.completed

↓

Router selects LogiCore Job Lifecycle Assistant

This ensures that the correct assistant responds to each situation.

Assistant Context Pack

Before an assistant acts, Titan Zero assembles a Context Pack containing relevant information.

Example context:

entity_type: job

entity_id: 4829

job_status: completed

customer_history: 14 previous visits

invoice_status: not created

site_type: residential

This context allows assistants to produce informed recommendations.

Assistant Decision Modes

Assistants operate in different decision modes depending on risk level.

Normal operations:

Assistant proposes action

Pulse executes

Higher-risk operations:

Assistant proposes action

Titan Zero governance evaluates

Action approved or rejected

This ensures that automation remains safe.

Assistant Memory

Assistants maintain contextual memory for lifecycle entities.

Memory types include:

short-term context

entity history
assistant interaction logs

Example:

job lifecycle history
customer service history
campaign performance history

Memory allows assistants to make better decisions over time.

Assistant Artifact Generation

Assistants may generate artifacts such as:

task plans
campaign plans
message drafts
inspection reports
financial recommendations

Artifacts are stored using the Titan Docs Engine.

Artifacts may be reviewed by operators before execution.

Assistant Governance

Assistant proposals are subject to governance rules.

Low-risk proposals may execute automatically.

Higher-risk proposals are reviewed by Titan Zero.

Example governance process:

Assistant proposal generated



Risk classification

↓
Titan Zero evaluation
↓
Decision issued

This ensures that assistants remain aligned with operational policies.

Assistant Interaction with Engines

Assistants do not interact directly with databases.

Instead, they interact through engine APIs.

Example workflow:

Assistant proposes invoice creation
↓
Titan Pulse executes action
↓
Money Engine API creates invoice
↓
Signal emitted: finance.invoice.created

This architecture maintains strict engine boundaries.

Assistant Monitoring

Assistants continuously monitor system signals.

Example signal streams:

work.*
finance.*
talk.*
customer.*
campaign.*

Assistants subscribe to specific signals relevant to their function.

Assistant Configuration

Assistants are configured using assistant profiles.

Example configuration fields:

assistant_name
assistant_type
monitored_signals
decision_rules
risk_threshold

These profiles define assistant behaviour.

Assistant Benefits

The assistant architecture provides:

proactive operational management

improved automation intelligence

better decision support

scalable AI integration

By combining assistants with signal-driven automation and governance oversight, TitanZero operates as an adaptive intelligent platform.